

1 System Architecture & Microstrip Implementation

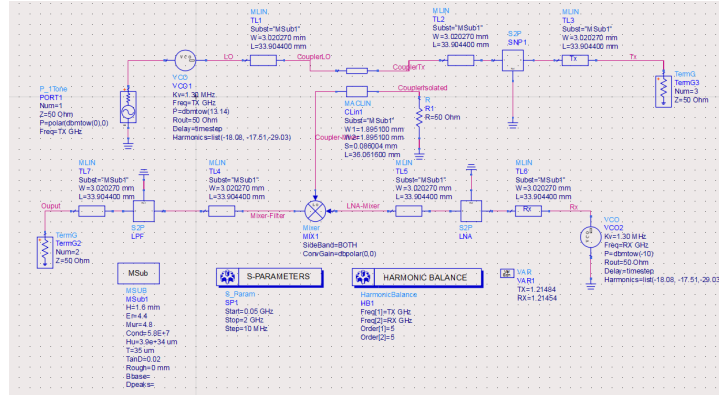
The RF transceiver for the S-band Continuous-Wave (CW) Doppler Radar was modeled and validated using Keysight Advanced Design System (ADS) based on PCB design principles for target detection [1]. The architecture integrates microstrip transmission lines (MLIN), a microstrip coupled-line directional coupler (MACLIN), S-parameter 2-port blocks (S2P), and a downconversion mixer (MIX1).

Table 1: Substrate Parameters (MSub1) and Microstrip Transmission Line Specifications

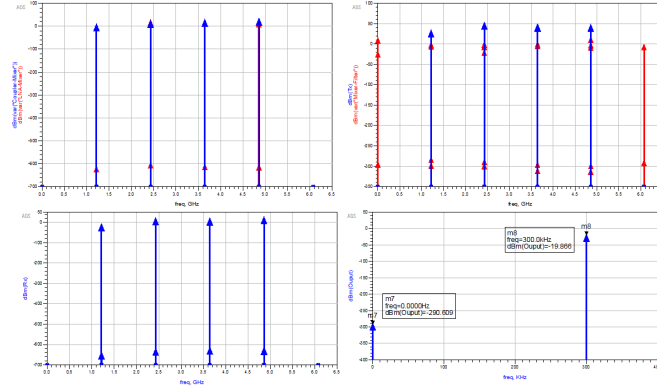
Substrate Parameter	Value	Microstrip Component	Dimensions / Parameters
Substrate Height (H)	1.6 mm	Standard Feedlines (TL1--TL7)	$W = 3.02027$ mm, $L = 33.9044$ mm
Relative Permittivity (ϵ_r)	4.4	Coupled Line Coupler (CLIN1)	$W_1 = W_2 = 1.8951$ mm, $S = 0.086$ mm, $L = 36.061$ mm
Conductor Thickness (T) / Cond.	$35 \mu\text{m} / 5.8 \times 10^7$ S/m	Transmit VCO (VC01)	$f_{TX} = 1.21484$ GHz, $P_{out} = 13.14$ dBm
Loss Tangent ($\tan \delta$)	0.02	Target Echo VCO (VC02)	$f_{RX} = 1.21454$ GHz, $P_{out} = -10.0$ dBm

2 ADS Schematic Capture & Downconverted Spectrum

System-level verification was performed using a two-tone Harmonic Balance (HB) simulation engine (HB1, Order[1]=5, Order[2]=5) to extract baseband Doppler frequency shifts from the downconverted signal spectrum [1].



(a) Keysight ADS System Schematic Capture



(b) Harmonic Balance RF Spectrum and Extracted IF Doppler Response

Figure 1: Keysight ADS CW Doppler Radar schematic capture and spectral response.

3 Doppler Frequency Extraction & Output Spectral Analysis

The transmitter carrier frequency is set at $f_{TX} = 1.21484$ GHz (13.14 dBm), while the target echo is synthesized at $f_{RX} = 1.21454$ GHz (−10.0 dBm). Downmixing extracts the target Doppler shift f_D :

$$f_D = |f_{TX} - f_{RX}| = |1.21484 \text{ GHz} - 1.21454 \text{ GHz}| = 300.0 \text{ kHz} \quad (1)$$

Table 2: Harmonic Balance Output Spectrum and Baseband Marker Summary

Marker Label	Frequency	Output Power (dBm)	Physical Description / Signal Component
Marker 7 (m7)	0.0000 Hz	−290.609 dBm	DC component (leakage/offset filtering)
Marker 8 (m8)	300.0 kHz	−19.866 dBm	**Extracted IF Doppler Signal (f_D)**
RF Spurious / Harmonics	1.2 to 4.9 GHz	> 0 dBm	TX Carrier, LO leakage, and upper harmonic terms

The baseband signal peak at 300.0 kHz (Marker 8 = -19.866 dBm) verifies successful homodyne downconversion and Doppler signature extraction.

References

- [1] Y. Suoglu and B. Kara, "Implementation of a Doppler Radar on PCB," Electronics Engineering, Sabancı University, Istanbul, Turkey.